ENTRI CODING

AI Glossary Guide for Beginners

Must-Know AI Terms

58+ Essential AI Terms Explained with Real-World Examples

How to use this guide

This glossary is built for absolute beginners starting their AI journey with Entri Coding Tamil. Each term includes a simple explanation and a real-world example or use case so the concept sticks. Read it top to bottom, or jump straight to a section you need before a class or interview.

1. Core AI Concepts

Artificial Intelligence (AI)

Meaning: The broad field of building machines and software that can perform tasks which normally need human intelligence, like understanding language, recognising images, or making decisions.

Example / Use case: *Google Maps predicting the fastest route, or a spam filter deciding which emails are junk.*

Machine Learning (ML)

Meaning: A branch of AI where a computer learns patterns from data instead of being explicitly programmed with rules.

Example / Use case: *Swiggy/Zomato predicting your delivery time based on past order data.*

Deep Learning (DL)

Meaning: A type of Machine Learning that uses multi-layered 'neural networks' to learn complex patterns, especially useful for images, speech, and text.

Example / Use case: *Face unlock on your phone recognising your face in different lighting.*

Neural Network

Meaning: A model inspired by the human brain, made of layers of connected 'nodes' that pass information forward to learn patterns from data.

Example / Use case: *The engine behind photo tagging apps that recognise your friends in pictures.*

Algorithm

Meaning: A step-by-step set of instructions or rules a computer follows to solve a problem or complete a task.

Example / Use case: *The algorithm YouTube uses to decide which video to recommend next.*

Artificial General Intelligence (AGI)

Meaning: A hypothetical, future stage of AI that can understand, learn, and perform any intellectual task a human can — not yet achieved today.

Example / Use case: *Still a research goal; today's AI (like ChatGPT) is 'narrow AI', good at specific tasks only.*

Turing Test

Meaning: A classic test proposed by Alan Turing where a human judges text conversations to see if they can tell a machine apart from a human.

Example / Use case: *Used historically to discuss how 'human-like' a chatbot's replies feel.*

2. Generative & Agentic AI

Generative AI (Gen AI)

Meaning: AI systems that create brand-new content — text, images, audio, code — instead of just analysing existing data.

Example / Use case: *ChatGPT writing an essay, or Midjourney creating a poster image from a text description.*

Agentic AI

Meaning: AI that can plan, make decisions, and take multiple actions on its own toward a goal, instead of only answering one question at a time.

Example / Use case: *An AI travel assistant that searches flights, compares prices, and books your ticket without you doing each step manually.*

AI Agent

Meaning: A single AI-powered program that can use tools (like search, calculators, or apps) and take actions to complete a task, often as part of an agentic system.

Example / Use case: *A customer-support agent bot that checks your order status in a database and replies automatically.*

Chatbot

Meaning: A program designed to hold a text or voice conversation with users, often to answer questions or provide support.

Example / Use case: *The 'chat with us' bubble on an e-commerce website answering FAQs.*

Multimodal AI

Meaning: AI that can understand and work with more than one type of input — text, images, audio, video — together.

Example / Use case: *Uploading a photo of a broken phone screen and asking an AI what the issue might be.*

Generative Adversarial Network (GAN)

Meaning: A pair of neural networks (a 'generator' and a 'critic') that compete with each other to produce increasingly realistic fake content, like images.

Example / Use case: *Creating realistic AI-generated human faces that don't belong to any real person.*

Diffusion Model

Meaning: A type of generative model that creates images by starting from random noise and gradually refining it into a clear picture based on a prompt.

Example / Use case: *Tools like Midjourney and Stable Diffusion generating artwork from a text prompt.*

3. Language Models & NLP

Large Language Model (LLM)

Meaning: A very large AI model trained on massive amounts of text so it can understand and generate human-like language.

Example / Use case: *Claude, ChatGPT, and Gemini are all LLMs used for writing, coding, and answering questions.*

Small Language Model (SLM)

Meaning: A lighter, smaller version of a language model, designed to run faster and cheaper, often on phones or laptops.

Example / Use case: *An offline AI keyboard that suggests the next word without needing the internet.*

Natural Language Processing (NLP)

Meaning: The field of AI focused on helping computers understand, interpret, and work with human language.

Example / Use case: *Google Translate converting Tamil text to English.*

Natural Language Generation (NLG)

Meaning: A part of NLP where AI produces human-readable text as output, often as a summary or response.

Example / Use case: *An AI tool auto-generating a product description from a list of features.*

Token

Meaning: A small chunk of text (a word, part of a word, or symbol) that an AI model reads and processes — the basic 'unit' of text for an LLM.

Example / Use case: *The word 'unbelievable' might be split into tokens like 'un', 'believ', 'able'.*

Tokenization

Meaning: The process of breaking text into tokens before feeding it into an AI model.

Example / Use case: *Happens automatically behind the scenes whenever you type a prompt into ChatGPT or Claude.*

Embedding

Meaning: A way of converting words, sentences, or images into a list of numbers (a vector) that captures their meaning, so similar meanings end up close together.

Example / Use case: *A search engine understanding that 'inexpensive laptop' and 'cheap laptop' mean nearly the same thing.*

Context Window

Meaning: The maximum amount of text (measured in tokens) an AI model can 'remember' and consider at once during a conversation.

Example / Use case: *If a chat gets very long, the AI may 'forget' the earliest messages once the context window is full.*

Hallucination

Meaning: When an AI confidently gives an answer that is factually wrong or made up, without realising it's incorrect.

Example / Use case: *An AI inventing a fake research paper title that sounds real but doesn't exist.*

Sentiment Analysis

Meaning: An NLP technique that determines whether text expresses a positive, negative, or neutral emotion.

Example / Use case: *A brand analysing thousands of product reviews to see if customers are happy or upset.*

Speech Recognition (ASR)

Meaning: Technology that converts spoken audio into written text.

Example / Use case: *Saying 'Hey Google, set an alarm for 7 AM' and having it typed and understood correctly.*

Text-to-Speech (TTS)

Meaning: Technology that converts written text into spoken audio.

Example / Use case: *Audiobook apps or GPS navigation voices reading directions aloud.*

Optical Character Recognition (OCR)

Meaning: Technology that reads text from images or scanned documents and converts it into editable digital text.

Example / Use case: *Scanning a printed Aadhaar card and extracting the name and number automatically.*

4. Prompting & Interaction

Prompt

Meaning: The input — a question, instruction, or piece of text — that a user gives to an AI model to get a response.

Example / Use case: *Typing 'Write a birthday wish for my friend in Tamil' into an AI chat.*

Prompt Engineering

Meaning: The skill of carefully wording and structuring prompts to get better, more accurate, or more useful responses from an AI model.

Example / Use case: *Adding 'explain like I'm a beginner, use a Swiggy analogy' to get a simpler answer from an AI tutor.*

System Prompt

Meaning: A hidden instruction set given to an AI at the start of a conversation to define its behaviour, tone, or role before the user even types anything.

Example / Use case: *A company setting up their chatbot with 'You are a polite support agent for XYZ store' behind the scenes.*

Zero-shot Learning

Meaning: When an AI performs a task correctly without being given any prior examples of that exact task.

Example / Use case: *Asking an AI to translate a sentence into a language it was never specifically trained/fine-tuned for that task, and it still gets it right.*

Few-shot Learning

Meaning: Giving an AI a small number of examples inside the prompt to guide it on how to respond.

Example / Use case: *Showing an AI 2-3 sample product titles before asking it to write a new one in the same style.*

Chain of Thought (CoT)

Meaning: A prompting technique where the AI is encouraged to reason step-by-step before giving a final answer, improving accuracy on complex problems.

Example / Use case: *Asking a math question with 'think step by step' so the AI shows its working before the final answer.*

Prompt Injection

Meaning: A security risk where hidden or malicious text tries to trick an AI into ignoring its original instructions.

Example / Use case: *A webpage secretly containing text like 'ignore previous instructions' to manipulate an AI browsing assistant.*

Retrieval-Augmented Generation (RAG)

Meaning: A technique where an AI first searches a knowledge base or documents for relevant facts, then uses that information to generate an accurate answer.

Example / Use case: *An internal company chatbot that looks up your HR policy PDF before answering a leave-related question.*

5. Training & Data

Training Data

Meaning: The large collection of examples (text, images, etc.) used to teach an AI model patterns during training.

Example / Use case: *Millions of labelled photos of cats and dogs used to train an image-classifying model.*

Dataset

Meaning: An organised collection of data used for training, testing, or evaluating an AI model.

Example / Use case: *A spreadsheet of past loan applications used to train a credit-risk prediction model.*

Data Labeling

Meaning: The process of tagging raw data with the correct answer or category so an AI can learn from it.

Example / Use case: *Marking thousands of photos as 'contains a car' or 'no car' before training a self-driving model.*

Pretraining

Meaning: The initial, large-scale training phase where a model learns general language or image patterns from huge amounts of data.

Example / Use case: *How an LLM first learns grammar, facts, and reasoning from billions of web pages before any fine-tuning.*

Fine-tuning

Meaning: Further training an already-pretrained model on a smaller, specific dataset so it becomes better at a particular task or domain.

Example / Use case: *Taking a general LLM and fine-tuning it on legal documents so it answers legal questions more accurately.*

Transfer Learning

Meaning: Reusing knowledge a model already learned from one task and applying it to a new, related task, saving time and data.

Example / Use case: *A model trained to recognise general objects being adapted to specifically detect defects on a factory line.*

Supervised Learning

Meaning: A training method where the model learns from data that already has correct answers (labels) attached.

Example / Use case: *Training a spam detector using emails already labelled 'spam' or 'not spam'.*

Unsupervised Learning

Meaning: A training method where the model finds patterns or groups in data that has no labels or predefined answers.

Example / Use case: *Grouping customers into segments based on shopping behaviour, without predefined categories.*

Reinforcement Learning (RL)

Meaning: A training method where an AI learns by trial and error, getting rewards for good actions and penalties for bad ones.

Example / Use case: *An AI learning to play a video game by getting points for winning moves and losing points for mistakes.*

Reinforcement Learning from Human Feedback (RLHF)

Meaning: A technique where human reviewers rate AI responses, and those ratings are used to further train the model to be more helpful and safe.

Example / Use case: *How chatbots like ChatGPT and Claude are fine-tuned to give more polite, accurate, and useful answers.*

Overfitting

Meaning: When a model learns the training data too closely (including its noise/errors) and performs poorly on new, unseen data.

Example / Use case: *A student who memorises answers for the exact practice questions but fails when the exam questions are phrased differently.*

Underfitting

Meaning: When a model is too simple to learn the real patterns in the data, so it performs poorly even on the training data itself.

Example / Use case: *A student who barely studied and can't answer even the basic practice questions correctly.*

Epoch

Meaning: One complete pass of the entire training dataset through the model during training.

Example / Use case: *If a model is trained for 10 epochs, it has seen the full dataset 10 times.*

Batch Size

Meaning: The number of training examples processed together before the model updates itself during training.

Example / Use case: *Feeding 32 images at a time into a model instead of one-by-one, to train faster and more stably.*

Model Weights

Meaning: The internal numerical values a model learns during training, which determine how it processes input to produce output.

Example / Use case: *When you 'download a model', you are mainly downloading its trained weights.*

Parameters

Meaning: The individual adjustable values (including weights) inside a model; more parameters generally means a bigger, more capable model.

Example / Use case: *You'll often see model sizes described as '7B' or '70B', meaning 7 billion or 70 billion parameters.*

Inference

Meaning: The process of using an already-trained model to make a prediction or generate a response on new input.

Example / Use case: *Every time you ask ChatGPT a question and it replies, that reply is generated during 'inference'.*

Graphics Processing Unit (GPU)

Meaning: Specialised computer hardware, originally for graphics/gaming, that is very efficient at the parallel calculations needed to train and run AI models.

Example / Use case: *Data centres using thousands of GPUs to train large models like GPT or Claude.*

6. Ethics, Access & Infrastructure

AI Bias

Meaning: When an AI model produces unfair or skewed results because its training data reflected existing human or societal biases.

Example / Use case: *A hiring AI unfairly favouring one gender because past hiring data was skewed that way.*

AI Ethics

Meaning: The study and practice of designing and using AI responsibly — covering fairness, privacy, safety, and transparency.

Example / Use case: *A company publishing guidelines on how their AI chatbot must avoid giving medical or legal advice.*

Explainable AI (XAI)

Meaning: AI designed so that its decisions can be understood and explained to humans, instead of being a mysterious 'black box'.

Example / Use case: *A loan-approval AI that can show which factors (income, credit score) led to a rejection.*

Vector Database

Meaning: A specialised database that stores embeddings (numerical representations of data) so an AI can quickly search for similar meanings, not just exact keyword matches.

Example / Use case: *Powers 'search by meaning' features, like finding similar support tickets even if worded differently.*

Application Programming Interface (API)

Meaning: A defined way for two pieces of software to talk to each other; for AI, it lets developers send prompts to a model and receive responses in their own app.

Example / Use case: *A food delivery app calling an AI API to auto-generate order summaries for restaurant partners.*